

Discrete Choice Model Interpretation & Business Recommendations

Estimated Time: 2-4 hours

Background

At Astrea, we use discrete choice models to help clients make data-driven portfolio decisions. A common challenge is answering "sourcing" questions: for example, if a product is discontinued, where will its sales volume go? Or, stated differently, understand the cannibalization effect of introducing a new product in a category.

Your task is to analyze the output of a simple discrete choice model, interpret its implications, and propose improvements to better address sourcing questions.

You'll be using an Multinomial Logit model, where the utility U for each item j , represented as:

$$U_j = \beta_0 + \beta_1 X_{1j} + \beta_2 X_{2j} + \dots + \beta_k X_{kj}$$

Where β_n is the n^{th} coefficient, and X_{nj} is the j^{th} item's n^{th} characteristic.

In order to estimate the β_n , we collected data from respondents that are exposed to trips (subsets of items) and have to declare which of those items they would choose.

Dataset & Model Output

You are provided with:

1. Two **CSV files** containing the results of a multinomial logit (MNL) model, one the model coefficients and the other, the respondents data to exposure to trips of items.
2. A short **Python script** to generate predictions.

Tasks

1. **Model Interpretation**
 - Examine the model coefficients and explain what they indicate about consumer preferences.
 - Identify any potential concerns or inconsistencies in the output (e.g., extreme coefficients, unrealistic substitution patterns).
2. **Business Implications**

- Suppose our client is considering discontinuing **Product X** (choose any item). Based on the model results, where do you expect its sales volume to shift?
- What additional analyses or model enhancements would you recommend to improve sourcing predictions?

3. **Technical Recommendations**

- What limitations does a standard MNL model have in answering sourcing questions?
- Suggest a potential improvement

Deliverable

Submit a brief **Jupyter Notebook or Python script** containing:

- Your analysis and interpretations in markdown/text cells.
- Any relevant exploratory code (e.g., visualizations, basic calculations).